

One Tick, One Link:

A Non-Autoregressive Challenge–Response Memory in Which Depth Is Bought with the World’s Silence

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Abstract

We report a sequence learner that is not autoregressive. It is *challenge–response*: one input per tick, one output per tick, and intervals in which the input is the identity and the model keeps emitting. Its two operations are inverses selected by the world — speech *binds* an observation into a superposed store, silence *unbinds* one link out of it — so that depth is neither a layer count nor a hyperparameter but the number of quiet ticks the world grants. Several partial responses mature in parallel and the strongest is said; what was said is heard back, and bound into the key of what the world did next. There is no reward, no label, and no gradient across a memory write. We verify in isolation that a chain is traversed one link per quiet tick and halts at its end, derive and confirm the store’s operating region $d > 2k \ln V$, show that binding is invertible only for unitary codebooks, and show that locality-sensitive addressing makes a near miss undetectable by construction. We measure without a single number: learners are ordered by the facts they maintain, codelength is read as an instrument against a control that cannot accumulate, and sources are screened with memoryless judges before a model sees them. On a synthetic continual stream binding, superposition and the event itself in the state are load-bearing; on 1.73M intensive-care measurements the memory accumulates where a bounded context cannot, and its one weakness against a counter is a rate, not a price. Components built on top of the mechanism — a routed operator graph, a read-back check, an exact per-context trace, a situation-bound episodic store — are reported with what they maintained and why they are left out; the last recalls a recurring situation across long gaps but not from one exposure. Every evaluation here is open-loop, so none tests the response half of the design. We argue that no static source can, set out what a world must have to test it — its own clock, influence without generation, recurring laws, persistent situations — and name where the design might be useful once it has been tested there.

1 Introduction

1.1 Three commitments of the autoregressive frame

The dominant frame for sequence learning factorises $p(x_{1:T}) = \prod_t p(x_t \mid x_{<t})$, feeds a prefix, and predicts the next symbol [Elman, 1990, Vaswani et al., 2017]. It has been extraordinarily productive, and it carries three commitments that are easy to stop seeing because everything around them assumes the same ones.

The objective is next-symbol prediction. Every metric derived from it inherits that shape. A system evaluated by per-symbol codelength is being asked how well it does the thing autoregression is for, whatever else it may have been built to do.

Computation per symbol is fixed. A forward pass costs what it costs whether the world gives the model a microsecond or an hour. Depth is a property of the architecture, chosen at design time. Recent stateful models [Gu and Dao, 2023] keep a running state rather than a window, but the computation per token is still fixed and the objective is still next-symbol.

The context is a prefix. An exact, recent, ordered window. Long-range association is bought by widening it [Beltagy et al., 2020, Dai et al., 2019] or by retrieving into it [Borgeaud et al., 2022, Weston et al., 2015]. Association at distance is thereby made a question about how much text you can hold, rather than about what you have stored.

For a system meant to run in a real temporal world — embodied [Brooks, 1991], or agentic over long horizons — all three sit badly. There is no prefix boundary in a room. Computation that cannot use extra time cannot exploit a lull, and lulls are most of what a real stream contains. And an exact window is the wrong idealisation for a background that in any organism is a fuzzy, multi-timescale trace rather than a transcript [Brainerd and Reyna, 2002].

1.2 What we put in their place

We take challenge–response as the primitive. There is no “a challenge arrives, now respond” — that phrasing is already the autoregressive shape, one prompt one completion. There is a stream, and every tick carries one input and produces one output. Some inputs are the world’s tokens; in the intervals between them the input is the identity and the model still emits, iterating on memory. The model’s own emissions are part of its context, interleaved with the world’s: distinguished from them, and combined with them.

The picture we have kept in view is two people talking. While the other is still speaking you have already begun to think; when they stop, or just as they stop, you begin to say something. That interval is the gap. What is forming in it is *not one trajectory* — several half-formed replies mature together and one of them is what comes out. Each clause of that sentence turned out to be load-bearing, and each had to be recovered separately after a run in which the mechanism measured inert (§9).

The mechanism follows from taking the identity input seriously. If speech means “there is something to combine”, then silence means “there is not”, and what is called for is the inverse operation. Binding writes; unbinding reads; the tick already says which. Nothing is scheduled, and — crucially — nothing needs to know whether a question is outstanding.

1.3 What is borrowed

Storage is content-addressed through a fixed map that no later learning can move, following the position argued in Yang [2026]: allocation buys freedom from interference and pays for it in retrieval difficulty, and the weakness of the family is that discrimination between addresses decays as addresses crowd. That trade organises our design, and one of our results is that in our implementation half of the crowding penalty was self-inflicted (§4.3).

The representation is holographic [Plate, 1995] in the vector symbolic tradition [Smolensky, 1990, Kanerva, 1988, 2009, Gayler, 2003, Eliasmith, 2013, Kleyko et al., 2023]: binding by circular convolution, reading by circular correlation, cleanup against a codebook. The readout is a delta rule [Widrow and Hoff, 1960, Rosenblatt, 1958]. The continual setting — regimes arriving, running and departing permanently, with no task identifier and no boundary signal — is the one in which catastrophic interference was characterised [McCloskey and Cohen, 1989, French, 1999] and to which complementary-systems and regularisation accounts respond [McClelland et al., 1995, Kirkpatrick et al., 2017, Parisi et al., 2019].

1.4 What this paper reports, and what it cannot

The mechanism of §3 is the one we stand behind. Several components were built on top of it, measured, and are kept in the implementation as switches that are off; §8 reports each with what it maintained and why it is not part of the mechanism. That is where a routed operator graph, a read-back check on the emission, an exact per-context trace and a situation-bound episodic store are discussed.

Every evaluation we have run is *open-loop*: the world’s next event is fixed in advance and does not depend on what the model said. That tests memory and prediction. It does not test the response half of challenge–response — what is said, and what the world does because it was said — and §10 argues that no static source can. The evaluation this design most needs is a world that runs on its own clock and that the learner’s behaviour influences without generating, such as a simulated physical environment; we set out what such a world must have and where the design might be useful once it has been tested there.

1.5 Contributions

1. A tick in which binding and unbinding are two halves of one loop selected by whether the world spoke, so that gap periods, self-output-as-context, and depth-from-time are consequences of a single rule rather than three mechanisms (§3, Figure 1).
2. An isolated demonstration that a chain is traversed one link per quiet tick, halts at its end, and is unharmed by surplus silence (§4, Figure 2).
3. The operating region of a superposed store, $d > 2k \ln V$, derived and confirmed, with the observation that circular correlation inverts convolution only for unitary codebooks — a Gaussian codebook loses almost half the signal with a single item stored (§4.2).
4. A structural result: locality-sensitive addressing makes a near miss undetectable *by construction*. Addressing needs same-content-same-bank and gains nothing from similar-content-similar-bank (§4.3).
5. A measurement method without a single scalar: learners ordered by the facts they maintain, prequential codelength read as an instrument and never as the objective, proper scores for answers that carry no information, and a screen that decides whether a source can show anything before a model sees it (§5).
6. Results on a synthetic continual stream and on a real clinical stream, including where the design loses and why (§6, §7), and a record of what was tried on top of the mechanism and not kept (§8).
7. An account of what static data cannot test, the properties a world must have to test the rest, and candidate applications (§10).

2 Philosophical commitments

These are not decoration. Each translates into a constraint that the implementation is checked against, and several of our negative results are consequences of holding to them.

2.1 Intelligence as the organisation of memory, not as optimisation against a scalar

The founding position of this work is that intelligence is an emergence of memory rather than a product of reward. The contrast is with the framing in which behaviour is whatever maximises

a scalar [Sutton and Barto, 2018, Silver et al., 2021]; here there is no reward, no value function, and no policy. It is closer to the tradition in which memory is the computational substrate rather than a subsystem attached to one [Hebb, 1949, Tulving, 1972, Gallistel and King, 2009].

The design consequence is sharp: the ledger may say what was surprising, and may therefore decide *what gets written and how deeply*, but it may not be descended. No gradient crosses a memory write. A node performs a write, not a fit. The readout’s delta rule is the one learner the design sanctions, and it is error-driven rather than frequency-driven (§2.2). Anything else fitted to the charge is a departure that has to argue for itself, and in §8 one such departure is reported and left out for that reason.

A test we apply before adopting any change follows from this: *would it still be right if the charge got worse?* A change justified only by a lower codelength is a change that has made the ledger the objective.

2.2 Compression as the currency, without frequency as the organiser

We charge prequential codelength [Dawid, 1984, Rissanen, 1978, Blier and Ollivier, 2018], in the tradition that identifies compression with understanding [Solomonoff, 1964, Hutter, 2005]. But we deliberately do not organise memory by frequency, and the reason is a claim about scale rather than about statistics.

Remark 1 (Why counts degrade with data). A per-location count distribution $N(o | u)/N(u)$ is a frequency estimate. As a stream accumulates, the count distribution at any location converges toward the corpus marginal — it becomes the global unigram. A counted prior therefore *stops discriminating exactly as the data grows*: it degrades under scale, which is the same shape of failure as address crowding.

More data tends toward disorder; what makes it mean something is organisation, and organisation need not be a Bayesian prior [Jaynes, 2003]. The delta rule already *is* an organiser of the right kind: it is error-driven, so a row holds what needed correcting here rather than what was common here, and a token already predicted stops updating. Counts accumulate and degrade under growth; the delta rule converges and is stable under it. There are therefore no counts and no escape chain in the mechanism, and the graceful degradation they used to provide is assigned to the dynamics rather than to a smoothing rule.

2.3 An imprecise background is the point, not a compromise

The background is a fuzzy multi-timescale trace. It is meant to say *which memories are relevant here*, not to reconstruct what was said. We state as a non-goal that precise context association is not what this system is for: a design that makes the background exact has rebuilt the prefix under another name, and inherits its costs. The nearest cognitive analogue is fuzzy-trace theory’s distinction between verbatim and gist representations [Brainerd and Reyna, 2002]; the nearest engineering analogue is that a multi-timescale reservoir is a context, not a transcript [Jaeger, 2001]. This commitment decided one of the explorations of §8: an exact per-context store answers a class of questions the mechanism cannot, and is left out because it is exactly what this commitment excludes.

2.4 Time as a resource, and the ledger that must not be gameable

If depth is bought with the world’s silence, then a system’s efficiency is part of its capability, and evaluation must be conducted where silences are real. A closely related commitment concerns the shape of the world. If the agent’s actions *generated* its observations, surprise could be minimised by finding a quiet corner — the dark-room objection to naive free-energy accounts [Friston, 2010, Friston et al., 2012, Clark, 2013]. The escape is not a patch but a property of reality: the world proceeds whether or not the agent attends, so behaviour *influences* what is

experienced without *generating* it, and missing something costs you later. No evaluation in this paper has that property; §10 returns to it.

2.5 No single number

A continual learner can be compared with another only by what it does, and what it does is many things. We order learners by inclusion of the facts they maintain — properties with known answers, each checked against a stated threshold — and report the order rather than a score. A faithful scalar exists only if that order is total, and the least number of scalars that represents it is its order dimension [Dushnik and Miller, 1941]; “give one number” asks for a choice the facts do not make. Where we report codelength, accuracy or AUROC, they are readings of particular instruments, never the verdict on a learner (§5).

2.6 Awe, and the limits of a constructed test

We take seriously that intelligence is not a quantity our sources can put under strain. A constructed stream can isolate mechanisms; it cannot decide whether those mechanisms amount to anything. A replayed real stream is closer to the world and still cannot answer back. We have tried to write the negative results as what they are — observations consistent with a plausible mechanism, on sources we argue cannot settle them — rather than as verdicts, and to say for each what it redirects us toward. Where the design should be judged is set out in §10, and we expect that judgement to be considerably harder than anything reported here.

3 Design

3.1 Notation and the algebra

Let d be the width, $\mathcal{C} = \{E_o\}_{o=1}^V \subset \mathbb{R}^d$ the fixed codebook, $\nu(\cdot)$ normalisation to the unit sphere, and $\otimes$ circular convolution,

$$(a \otimes b)_k = \sum_{j=0}^{d-1} a_j b_{(k-j) \bmod d}. \quad (1)$$

Write a^* for the involution $a_0^* = a_0$, $a_j^* = a_{d-j}$, and $m \odot a = m \otimes a^*$ for circular correlation.

Definition 1 (Unitary codebook). $a \in \mathbb{R}^d$ is *unitary* if $|\hat{a}_j| = 1$ for every Fourier coefficient $\hat{a}_j$. We construct $\mathcal{C}$ by drawing phases uniformly and taking an inverse transform, so every E_o is unitary.

Proposition 1 (Exact inversion). *For unitary a , $a \otimes a^* = \delta$, hence $(a \otimes b) \odot a = b$ exactly. For general a , $(a \otimes b) \odot a = b \otimes (a \otimes a^*)$, and $a \otimes a^*$ is a δ only in expectation.*

Sketch. Convolution is multiplication in the Fourier domain and the involution conjugates, so $\widehat{a \otimes a^*}_j = |\hat{a}_j|^2$. This is identically one exactly when a is unitary; for Gaussian a the $|\hat{a}_j|^2$ are χ^2 -distributed with substantial variance, and the reconstruction is degraded before any interference. $\square$

Proposition 1 is not a technicality. Table 1 measures the cost: with a *single* triple stored and no interference whatever, a Gaussian codebook reconstructs at cosine 0.527.

3.2 The tick

Write x_{-1}, x_{-2} for the two previous observed tokens, c_i for the state of the i -th response, τ_i for the codebook identity it has resolved to, $w_i \in [0, 1]$ for how live it is, and $h(\cdot, \cdot)$ for a hash of two token identities into $\{1, \dots, B\}$.

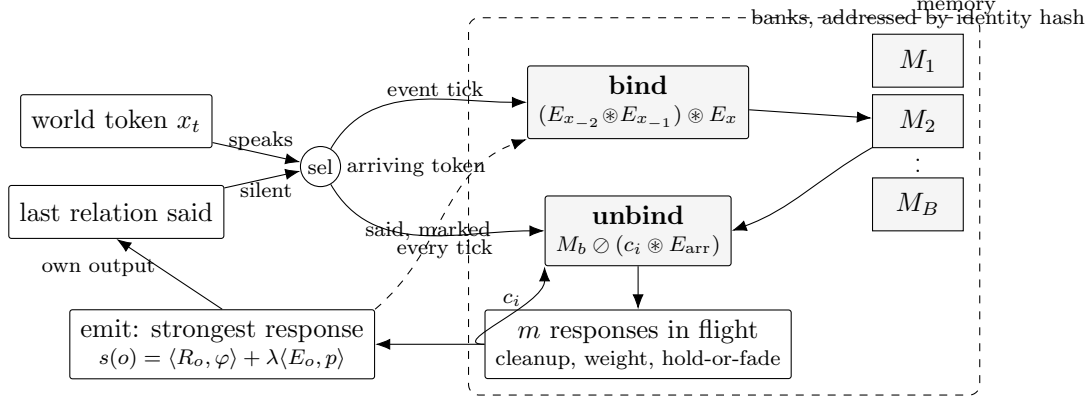


Figure 1: One tick. A token always arrives: the world’s when it speaks, the last relation it said when it does not. Speech binds a triple into the bank named by the two identities; every tick each response in flight unbinds one link out of the bank named by its own identity and the arriving one. The emission is the most matured response, scored by a learned row plus a direct codebook comparison, and it re-enters as the model’s own context. What was said also enters the key of what the world did next, marked as said (dashed). Nothing in the loop needs to know whether a question is outstanding.

Speech binds.

$$\mathcal{M}_{h(\tau_{x_{-2}}, \tau_{x_{-1}})} \leftarrow \nu((E_{x_{-2}} \otimes E_{x_{-1}}) \otimes E_{x_t}). \quad (2)$$

For a presented link $x r y$ the triple formed at the y tick is exactly $(E_x \otimes E_r) \otimes E_y$, and no episode boundary was needed to know that. The sum is *not* renormalised: $\mathcal{M} \leftarrow \nu(\mathcal{M} + t)$ shrinks everything already stored by $1/\sqrt{2}$ per write, giving the first item weight $2^{-k/2}$ — a sliding window, which is the one thing a superposition must not be.

Silence unbinds; and so does speech. On *every* tick each response advances one link:

$$q_i = \nu(c_i \otimes E_{arr}), \quad r_i = \nu(\mathcal{M}_{h(\tau_i, \tau_{arr})} \oslash q_i), \quad (3)$$

$$\tau'_i = \arg \max_{o \in \mathcal{C}} \langle E_o, r_i \rangle, \quad c_i \leftarrow E_{\tau'_i} \text{ iff } \langle E_{\tau'_i}, r_i \rangle > \theta(d, V). \quad (4)$$

Cleanup ranges over the whole codebook: naming a token cannot require that it has been learned before. Because $\otimes$ commutes, a response that has just reached a_1 after hearing r carries $E_{a_1} \otimes E_r$ as its next key with no bookkeeping. This is where the model’s own output does its work: the response *is* the position reached, and the relation stays on the table as the operator.

Arrival holds; speech supersedes.

$$w_i \leftarrow \begin{cases} 1 & \text{cleanup accepted,} \\ \gamma w_i & \text{rejected and the response has never advanced,} \\ w_i & \text{rejected and it has advanced at least once (it has arrived),} \end{cases} \quad (5)$$

and on every event tick $w_i \leftarrow \gamma w_i$ for all i before a new response is seeded at the observed token. From inside, “the chain ended” and “the chain never started” are identical; the only thing separating them is whether anything was ever retrieved. Holding a half-formed reply while the other person is quiet, and dropping it when they say something new, is the same picture.

The emission reads the most matured response. Let $i^* = \arg \max_i (w_i, \text{recency}_i)$ lexicographically. Then $p = c_{i^*}$, and the state is pulled toward the background on every

read, $p \leftarrow \nu(p + \alpha \sum_k b_k)$ with b_k the background bands of §3.5. The features are $\varphi = [p; E_x; \text{bound traces}; \text{event-locked bands}]$, where the bound traces are $E_{x-j} \circledast E_x$ for the two previous events and $b_k \circledast E_x$ for each band, and

$$s(o) = \underbrace{\langle R_o, \varphi \rangle}_{\text{learned}} + \lambda \underbrace{\langle E_o, p \rangle}_{\text{cleanup}}. \quad (6)$$

The event itself, E_x , is the lag-zero member of the family the bound traces form, since $E_x \circledast \delta = E_x$; without it a conjunction is present without its parts, and a new pair of familiar parts cannot be answered from the parts. Scores are normalised by subtracting their maximum, never by clamping them: the rows are unregularised and grow, and a clamp makes every large score equal.

Remark 2 (Why not the mixture). Averaging unit responses and renormalising divides each by $\|\sum_i w_i c_i\| \approx \sqrt{\sum_i w_i^2}$. Measured, the response holding the answer arrived at the readout at 0.24–0.28 against a codebook floor of 0.255: present and indistinguishable from noise. “Several half-formed answers mature together and one of them is said” is not “say their average”. Reading the strongest is the natural readout of parallel responses, not a competition between mechanisms.

The write. Rows are updated by a delta rule

$$R_o += \eta (\mathcal{K}[o = x] - q(o)) \varphi \quad \text{for every } o, \quad (7)$$

against the emitted distribution q — the distribution the ledger charged. The negatives are therefore what the model would actually have said, weighted by how much it would have said them. Truncating them to the few tokens ranked highest looks like the same principle and is not: on a stream whose right charge for a token of probability p is exactly $-\log_2 p$, sixteen negatives leave tokens rarer than 2^{-10} with a probability near 2^{-50} (§5). The step η is fixed.

3.3 Three streams

The model hears itself. Three self channels are fed back every tick, each through its own fixed rotation so that they do not superpose into the world’s channel, and each only into the fastest band of the background (A5 below):

- *covert* — what it currently takes the answer to be, emitted every tick because the system is always outputting;
- *overt* — what it has said out loud, once the evidence stops moving;
- *write* — which memory it is touching: a fixed vector for the bank the response last unbound.

These streams are context, not content: they reach the features and the fastest band, and nothing they carry is ever stored as having happened.

What was said reaches memory once, in the key and never in the value. When the world says x after the model said a (and $x \neq a$), a second triple is written under the pair (the world’s token before a , a marked as said), with value x :

$$\mathcal{M}_{h(\tau_{x-1}, \hat{\tau}_a)} += \nu((E_{x-1} \circledast (S \circledast E_a)) \circledast E_x), \quad (8)$$

where S is a fixed unitary vector and $\hat{\tau}_a$ an identity in a space of its own. The mark matters. The silent read above uses the last thing said as its relation, so an unmarked (x_{-1}, a) is exactly the key a response at x_{-1} reads: a finished walk then read “what the world did after I said a here” and walked off its answer. An act is heard back on its own channel, not as the world’s word. Nothing in the mechanism reads these keys yet; they are what a world that reacts would be learned from (§10).

3.4 Structural constraints

Five constraints are asserted by unit tests that run on every build.

- A1 Rate limit.** Constant computation per tick. The only way to think longer is to occupy more ticks, and ticks are given by the world.
- A2 Event-driven accounting.** Charged only when the world speaks, so “the identity input injects no new content” is a consequence of the ledger rather than an added axiom.
- A3 Read/write separation.** The write is content addressed and deterministic; reads may wander.
- A4 No content write at baseline.** Identity ticks write nothing, asserted bit-for-bit over 500 baseline ticks.
- A5 Timescale reachability asymmetry.** The world writes all rungs; the model’s own output reaches only the fastest, so self-generated content contributes exactly zero to the slow timescales — a structural ceiling on confabulation. What was said enters memory only as a marked key, never as a stored value.

3.5 The background, and why its key is frozen

The background is a cascaded band-pass ladder: one use of a ladder, on the time axis only. Three rungs span one to 256 ticks log-uniformly. The memory itself is flat, because a ladder inside the store would be an abuse of the device.

The bands are snapshotted *at the last event*, so the retrieval key stops moving when the world does. The reason is that the bands carry two things at once: which regime this is, which must hold across a silence, and how long the world has been quiet, which changes every tick. With both in the key, elapsed time decides whether a memory fires at all — measured, a fact learned at six ticks of silence did not answer at one, scoring 0.000 where the same pair presented as an ordinary fact scored 0.74. A clock in the key is a bug; the live fast rung remains available to the commit rule, which is where elapsed time belongs.

3.6 What is not in the mechanism

There is no routed operator graph: an earlier design passed the state through a learned transform chosen by content from a small-world graph. Its memory is not inert, but on every reading we have, removing it costs nothing and keeping it costs something (§8). There is no count, no escape chain, no reward, no task boundary and no replay buffer. Further components exist in the implementation as switches that are off, each reported in §8: a read-back check on the emission, a gate on silent hops that return to the token just heard, a stream reporting where the last write went, a per-context exact trace, a situation-bound episodic store, and step-size rules.

4 Isolated verification

The evidence in this section is from unit tests on synthetic memories in which every quantity is known. It says what the mechanism does. It says nothing about whether the mechanism is useful on any data. We keep the two kinds of evidence in separate sections throughout, because conflating them is the easiest way to overstate a result of this kind. Every claim here is asserted by a named test that passes on the mechanism as described.

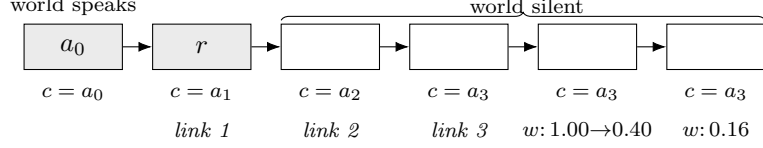


Figure 2: A walk $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow a_3$ presented forty times among unrelated traffic, then asked with the start and the relation. One link per tick; the end of the chain has no outgoing link, so the unbinding returns noise, cleanup refuses it, and the response holds its answer rather than stepping past it. Surplus thinking time is harmless. The relation’s own response fades on the same rule.

4.1 One link per quiet tick

Four things had to be right simultaneously for Figure 2, and each was found by a four-second test rather than by an end-to-end run (§9): the relation is the operator and the model’s own output is the position, not the reverse; seeding must retire the *least live* response rather than the oldest, because on ties “oldest” is the slot holding the response that has just advanced; responses must be weighted by whether they are still advancing; and the configuration must lie inside the operating region below. The same assertion is what caught the unmarked form of Equation (8): with what was said in the world’s key space, the cursor on the answer fell from 0.89 to 0.03 at the first quiet tick.

4.2 Capacity, and the operating region

Proposition 2 (Operating region). *Let $\mathcal{M} = \sum_{i \leq k} t_i$ with t_i unit and mutually near-orthogonal, and let cleanup take the best match among V codebook entries in $\mathbb{R}^d$. Then a stored triple returns $\langle \hat{\mathcal{M}}, t_j \rangle \approx k^{-1/2}$, an unwritten key returns $\approx d^{-1/2}$, and a random vector’s best codebook match is $\approx \sqrt{2 \ln V / d}$. Retrieval is usable when*

$$\frac{1}{\sqrt{k}} > \sqrt{\frac{2 \ln V}{d}} \iff d > 2k \ln V. \quad (9)$$

At $d = 256$ we measure cosine 0.043 with $k = 401$ against a predicted 0.050. Equation (9) is what gives content addressing a job that is not about accuracy: the address chooses *which* superposition to unbind, each bank stays under capacity, and the read stays one unbinding however much has been stored. Related capacity analyses for distributed representations appear in Plate [1995] and Frady et al. [2018].

It also fixes the cleanup threshold, which must not be a hand-set constant: $\theta(d, V) = \kappa \sqrt{2 \ln V / d}$. A hand-set 0.25 once sat against a floor of 0.452 — *below* the noise — so cleanup accepted noise on 97.6% of silent ticks and the response was randomised every tick while the instrument reported near-perfect success.

codebook	$k = 1$	$k = 16$	$k = 64$
Gaussian, unit norm	0.527	0.205	0.120
unitary (random phase)	1.000	0.251	0.127
$k^{-1/2}$ (theory)	1.000	0.250	0.125

Table 1: Reconstruction cosine, $d = 256$, mean of 12 draws. Proposition 1 in numbers: a Gaussian codebook loses almost half the signal with a *single* item stored, and Proposition 2 is a statement about unitary vectors.

4.3 Addressing must not preserve locality

A key nobody wrote still hashes into an occupied bank, and its occupants answer. The end of a chain therefore does not retrieve *nothing*; it retrieves a neighbour. No threshold separates the two, because both are real stored content of comparable strength. This is address crowding [Yang, 2026] in its sharpest form.

A read-back check was built for it — bind the candidate back onto the key and ask whether that triple is in the bank at all — and it failed instructively.

Lemma 1 (Locality defeats verification). *Under sign-bit locality-sensitive hashing [Indyk and Motwani, 1998, Charikar, 2002], the occupants of a bank have keys close to the query. If cleanup returns some occupant’s target y_i then, for unitary vectors,*

$$\langle \nu(p_i \otimes y_i), \nu(q \otimes y_i) \rangle = \langle p_i, q \rangle,$$

and $\langle p_i, q \rangle$ being large is precisely why p_i and q share a bank. The verification is defeated by the property it was built to detect.

Addressing needs same-content-same-bank and gains nothing from similar-content-similar-bank; the second property is pure cost and it is what makes a near miss undetectable. Scrambling the code removes the locality but makes the address brittle, since write and read reach the same convolution by different floating-point paths and one flipped sign then means an unrelated bank. That dilemma is false: cleanup has already resolved the response to an exact codebook token and the write holds both identities, so the address can be a hash of the *identities*. Vectors carry the superposition; identities carry the address. With that in place the read-back check becomes inert on the walk — zero rejections. Its later use on the emission is reported in §8.

5 How we measure without a single number

A design with no objective still has to be measured, and every instrument we used can be mistaken for an objective. This section fixes what each instrument is for before any result is reported.

5.1 Codelength is an instrument

We charge prequential codelength on the ticks where the world speaks: predict, pay $-\log_2 q(x)$, then write [Dawid, 1984, Blier and Ollivier, 2018]. It is the currency of §2.2, and it is a reading taken inside a counter’s frame — the codelength of the next token, which is the autoregressive target this design refuses. It is useful for one thing: a margin that moves when a mechanism is repaired says the repair reached something real. We report PPM [Cleary and Witten, 1984] beside it not as a standard to beat but as a control that *cannot accumulate*: a bounded context carries nothing across itself, so a sustained gap is evidence of accumulation. A number maximised is a reward, so no change is adopted because it lowers codelength alone.

5.2 Known answers for the instruments themselves

Several of our worst errors were in instruments, not mechanisms (§9). An instrument is therefore first run where its answer is known. Two examples recur below.

- *Rarity*. On an i.i.d. Zipf stream the right charge for a token of probability p is exactly $-\log_2 p$, so the excess over it, split by rarity, must be flat. A readout that writes off what it has seen rarely shows as excess growing with rarity. Writing against sixteen negatives gave an excess of 57.0 bits on tokens rarer than 2^{-10} ; writing against the whole emitted distribution, 0.47.

- *Comparators.* An online counting comparator first scored below chance on a clinical outcome, because a feature with no variance yet observed was standardised against a scale floor of 10^{-6} . A planted linear signal now has to be found (AUROC above 0.8) before the comparator is used.

5.3 Facts, not a score

Learners are compared by the facts they maintain (Table 2). Each fact has a known answer on a stream built for it and a stated threshold; a learner either maintains it or does not, and the measured value is reported beside the verdict. Learners are then ordered by inclusion: $a \geq b$ when a maintains every fact b maintains. We report the incomparable pairs, the width of the order and bounds on its dimension; where the order is not total, no single number is faithful to it.

	fact	threshold (truth)
F1	learns a deterministic successor	charge < 0.5 bits (0)
F2	is not confidently wrong on noise	i.i.d. charge below uniform
F3	does not write off the rare	rare minus common excess < 1 bit (0)
F4	the state carries the previous event	decode > 0.5 (chance 0.25)
F5	the state carries the silence	decode > 0.5 (chance 0.25)
F6	judges presence from history	AUROC > 0.9 (chance 0.5)
F7	judges order from history	AUROC > 0.8 (chance 0.5)
F8	answers a new pair from its seen part	novel charge < 0.25 bits (0)
F9	silence does not worsen an ungrounded answer	Brier after $\leq$ at ask +0.01
F10	the walk alone answers a relation	rows off: charge below uniform
F11	says back a key’s value set this turn	accuracy > 0.5 of 64 (chance 0.015)

Table 2: The facts. Each is measured on a stream whose answer is fixed by construction. F6, F7 and F9 use a stateful judge: an episode of items, a question about the episode’s history, a silence, and the answer spoken by the world; with the episode wiped before the question it is at chance.

5.4 Proper scores for answers with nothing under them

An answer given with no information behind it is correct about half the time at a 0.5 threshold whatever it does, so accuracy on it is noise. We score such answers with the Brier score and expected calibration error [Brier, 1950]. The distinction decided one exploration: read by accuracy, the silence after a question appeared to make an ungrounded answer worse, and a gate was built to stop it; read by proper scores, the silence makes it *better*, and the gate was withdrawn (F9; §8).

5.5 Screening a source before a model sees it

If the last few events, counts, or counts with elapsed time already answer a question, no state can show anything on that question, however good it is. So before a source is used we score memoryless judges on it — last- k events, bags of counts, counts with elapsed time, and for a retrieval question the rule that answers it by lookup — and use the source only if the question needs what they lack. Three sources failed or passed this screen in instructive ways (Appendix A).

5.6 One probe for both arms, and comparators in the same regime

Two mechanisms compared through a readout must be compared through the same readout, with the same regularisation sweep, on the same items. And a comparator must learn in the

same regime as the model — online, one pass, each item predicted before it is learned from — or the comparison measures the regime. Both rules were learned by breaking them (§9).

6 A synthetic continual stream

The evidence in this section is from the full system on a stream we built: 90k ticks, 12 regimes, two seeds, each figure the mean over both. It is source-dependent.

6.1 The source

Regimes arrive, run, and depart permanently; no task identifier, no boundary signal, no reset. Each carries first-order facts, a Latin square, a product code, composition chains, and walks. Two families can only be answered by computing: composition queries are posed *once* per chain and withheld Latin cells are asked once each, so nothing about them can be memorised from a previous presentation.

The generator is self-tested before any model is built, because on this project it produced as many serious defects as the model did.

quantity	Latin square	product code
$I(T; A \mid D), I(T; B \mid D)$ (bits)	0.126 / 0.127	non-zero by design
prequential single-cue baseline	0.201 / 0.195	0.486 / 0.402
analytic value	$1/5 = 0.200$	0.500
composition leakage		0

Table 3: Source self-check, measured on the emitted stream with no threshold. Marginal information uses the Miller–Madow correction [Miller, 1955]. The single-cue baseline is *prequential* — predict from counts so far, then update — because an in-sample ceiling is an oracle that no model charged before it sees the answer can reach.

Both Latin cues are drawn uniformly. With one cue Zipf-distributed and the other uniform, $P(T \mid A)$ is uniform exactly when p_B is, so such a square is not zero-marginal: a predictor using one cue alone reaches 0.27. No choice of square repairs it, and making both Zipf only makes the leak symmetric; the frequency skew therefore lives outside the square, where it forges no shortcut. The price of this symmetry is recorded in §9: a source with uniform marginals hid a defect in the emission path for as long as it was the only source.

6.2 What the mechanism does on it

Table 4 reads as the design says it should. Binding is load-bearing: without it every family collapses toward its single-cue baseline. The superposed store is load-bearing: without it Latin falls from 0.49 to 0.43 and the walk’s support for the answer from 0.38 to 0.31. Bank count matters until each bank is inside the operating region of Proposition 2 and not after (65536 banks read like 8192). The event itself in φ carries retention: without it retention falls from 0.50 to 0.31. Two things the source does not reward: removing the walk through the gap changes little here, and the composition queries posed once are not answered by any arm (0.000). Chaining two stored facts on first asking is where the design most needed to show something on this source, and it does not.

6.3 The step size, and what this source rewards

A step of 0.5 is better than 0.2 on every reading of this source (Table 5). It is not the step of the mechanism, because the same step is confidently wrong on noise: on an i.i.d. stream it charges

arm	Latin	product	retention	answer bits	walk support
the mechanism	0.486	0.765	0.498	3.98	0.379
bind off (<i>must collapse</i>)	0.063	0.505	0.200	6.04	0.012
no superposition	0.427	0.733	0.434	4.18	0.307
one bank	0.429	0.751	0.487	4.11	0.315
256 banks	0.454	0.762	0.494	4.10	0.337
65536 banks	0.487	0.765	0.494	3.98	0.375
no self block	0.458	0.712	0.305	4.42	0.287
no walk through the gap	0.484	0.774	0.519	3.99	0.392

Table 4: Two seeds, codelength charged at the answer tick only. Against the single-cue baselines of Table 3 (0.20 and 0.49) both conjunction families are answered well above a predictor that ignores half the challenge. The collapse arm holds, which is what makes the rest readable. The walk supports the answer when superposition is on and there are enough banks to keep each under capacity; without superposition every measure falls.

arm	Latin	product	retention	answer bits	walk support
the mechanism, step 0.2	0.486	0.765	0.498	3.98	0.379
step 0.5	0.611	0.844	0.640	2.85	0.570
sixteen negatives	0.490	0.764	0.471	4.02	0.343

Table 5: Two seeds. A larger step is better on every reading of this source. Which negatives the write uses is not what matters here.

6.73 bits against a uniform 6.00 and fails F2, and on a label flipped one time in five it chases the flips (§6.6). The synthetic source is deterministic inside every regime, so it rewards exactly the step that fails wherever the target is noisy. One global step cannot serve both, and we choose the one that does not fabricate on noise. What would separate learning structure from chasing noise is a quantity other than the error, and §11 records that we do not have it.

The same table shows what the choice of negatives does and does not do. Writing against sixteen negatives instead of the whole emitted distribution leaves the synthetic readings where they are; on a skewed stream it writes off rare tokens (§5). The source cannot see the difference, which is why the whole distribution was chosen on a stream that can.

6.4 What this source cannot settle

arm	Latin	product	retention	answer bits	walk support
the mechanism (3 rungs)	0.486	0.765	0.498	3.98	0.379
all self streams off	0.489	0.778	0.549	3.83	0.439
covert stream off	0.486	0.775	0.543	3.87	0.430
live bands in the key	0.492	0.757	0.462	3.94	0.365
1 rung	0.544	0.823	0.728	4.87	0.575
2 rungs	0.469	0.737	0.492	4.16	0.355
5 rungs	0.467	0.767	0.506	3.87	0.391

Table 6: Two seeds. Components that read net-negative or ambiguous on this source.

Two components read net-negative here: the model’s own streams fed back into the background, and background depth beyond one rung (Table 6). One mechanism explains both. φ is a fixed-width retrieval key; anything added to it that is not a function of the fact being asked about adds variance without signal. On this source an answer is determined by its cues, so

what the model just said carries no information about it while making φ differ between two presentations of the same fact; slow bands carry which regime this is, which the cues already say. A single rung retains most (0.73) because it forgets fastest what does not matter here, and charges most at the answer (4.87 bits) for the same reason.

We rule on neither. Both explanations rest on the source’s answers being cue-determined, which is the property of a constructed source most open to doubt; the hypothesis predicts a sign flip where the model’s own trajectory carries necessary information, and no source here has that property. The self streams are the model hearing itself, which is the design’s first commitment about context; the evidence against them is a source on which hearing oneself cannot help.

6.5 A question about history, and the silence after it

The same mechanism was asked questions whose answer depends on the order of what came before: an episode of items, a question (was this item in the episode; which of two came later), a silence, and the answer spoken by the world. The judge is stateful by construction: with the episode wiped just before the question it sits at chance.

question	accuracy	AUROC	ECE	PPM accuracy
presence	1.000 / 0.999	1.000 / 1.000	0.005 / 0.006	0.738 / 0.743
recency	0.956 / 0.954	0.991 / 0.990	0.010 / 0.013	0.732 / 0.731

Table 7: Two seeds (7 / 8). PPM sees the same stream and cannot hold the episode across its bounded context.

What the silence does to an answer with nothing under it is read by proper scores (§5). With the episode wiped, the Brier score from the ask to the end of the silence falls from 0.336/0.329 to 0.280/0.275 on presence and from 0.442/0.415 to 0.309/0.302 on recency; calibration error halves (0.313 $\rightarrow$ 0.165 and 0.423 $\rightarrow$ 0.205 on seed 7). The walk commits to an answer during the silence in 99.7–100% of questions, wiped or not. The silence does not fabricate: on an ungrounded answer it moves the model toward the base rate.

6.6 Noise floors

A fixed-step least-mean-squares readout has a steady-state misadjustment proportional to the step and to the noise on the target, and none on a deterministic target [Widrow and Hoff, 1960]. This is checked on a question with a known ceiling: the spoken answer is flipped with probability 0.2, so the best AUROC is exactly 0.800 and the best Brier score exactly 0.160.

step	AUROC	Brier	ECE
0.2 (the mechanism)	0.771 / 0.774	0.206 / 0.203	0.103 / 0.094
0.05	0.798 / 0.791	0.177 / 0.179	0.073 / 0.074
0.02	0.805 / 0.793	0.169 / 0.171	0.053 / 0.057
0.005	0.807 / 0.788	0.172 / 0.179	0.049 / 0.071

Table 8: Two seeds. The ceiling is AUROC 0.800, Brier 0.160; figures above 0.800 are sampling error on 3000 read questions.

The alternative reading, that the answer is simply too rare to learn, is rejected: with the question held easy, accuracy stays at 0.999–1.000 as the answer grows rarer, from one answer in 10 events to one in 514.

The converse error is also available and was once made: a residual on a deterministic stream read as this misadjustment. On a stream whose map is replaced every 10^5 events, the residual

in the last tenth before a change *shrinks* as the step grows (0.024 at step 0.5, 0.037 at 0.2, 0.085 at 0.05), whereas a misadjustment grows with it, and recovery after the change is slower at the smaller step (9768 against 20486 events). On a stationary stream of 6×10^5 events the residual is still falling in the last segment at every step ($0.0080 \rightarrow 0.0065$ at 0.2). It is a descent in progress, not a floor.

Two rules that shrink the step as a row settles were measured on the same changing stream. A step of $\eta/(1+n)$ never learns (5.17 bits in the first segment, 4.81 in the tenth, against 0.97 and 0.04 for the fixed step). An error-driven step learns ($4.47 \rightarrow 0.58$) but converges an order of magnitude worse than the fixed step and recovers no better (6.23 bits after the change, against 1.38). Neither can tell “the error is small because this row has learned” from “the error is small because it is nearly there”.

7 A real stream: intensive-care measurements

This section is not a benchmark claim. It records what one real, irregularly timed stream says about the mechanism, and where it cannot say anything. Every figure is from the mechanism as described; the clinical runs are a single pass over a fixed data set, so seed variation enters only through the codebook and is small beside the effects reported.

7.1 Data, and why this source

PhysioNet/CinC Challenge 2012 [Silva et al., 2012], set-a: 3996 intensive-care stays, 1.73M timestamped measurements. Each measurement is one token (parameter, octile of its value), $V = 296$. Elapsed time between measurements enters as idle ticks, $\lfloor \log_2 \rfloor$ of the gap in minutes, capped at 11, so the world’s silence is the world’s own clock. In-hospital death: 554 stays (13.9%). The source passed the screen of §5: the observation times carry information of their own (Appendix A). Memory runs continually across all stays and is never reset; only the volatile state — responses in flight, the background cascade — restarts per stay.

The unit of prediction had to be chosen, and the choice decides what is measured. At the granularity of a panel (the set of parameters recorded at one time) the gap appeared to carry 0.485 bits about the next panel, yet conditioning PPM’s context on the gap *cost* it 0.047 bits per event, because 3600 distinct panels fragment every context it needs. At the granularity of a single parameter the same conditioning gains PPM 0.18 bits. An advantage that appears under one individuation and not under a better-posed one is not an advantage of the learner, so we use one token per measurement.

7.2 Codelength, as an instrument

Over the whole stream the mechanism charges 3.319 bits per event against 4.097 for PPM at its best order (order 2), a margin of 0.78. The margin appears in the first decile and holds: 0.73 in the first tenth of the seen contexts, 0.76–0.83 in every later one. On events whose order-2 context the counter has never seen, the margin is +0.26 (5.57 against 5.83 bits). This reading is taken inside a counter’s frame, and §5 states what it is for: a control that cannot accumulate, against a memory that does.

Where the margin is negative, and why. Take the events whose order-2 context is new but whose order-1 context is not, and split them by how often this exact (order-1 context, target) association had occurred before (Table 9). A counter holds an association after one occurrence; the delta rule needs several corrections to move the same mass. The negative band is a rate, not a price: it is where the memory has not yet arrived.

association seen before	events	ours	PPM-C o2	margin
never	3536	8.07	10.84	+2.77
once	1841	7.29	6.32	−0.97
2–4 times	2606	6.39	5.58	−0.80
5–19	4096	5.04	4.88	−0.16
20 or more	10365	3.63	3.64	+0.01

Table 9: The one cell that stays negative. Bits per event.

Composition or rarity. A new context is mostly made of rare tokens, so its loss mixes a failure to compose with rarity. Read at matched rarity of the target: for targets seen a thousand times or more, the margin is +0.83 in seen contexts and +0.52 in new ones; for a hundred to a thousand, +0.52 and +0.27; for ten to a hundred, +0.17 and −0.50. A new context still costs something at matched rarity, and the cost grows as the target gets rarer. That is the delta rule’s rate again, now compounded with composition, and it is the reading the rarity instrument of §5 makes possible: before the write was made dense, the same column was dominated by rare targets written off.

It does not separate organisation from counting. Take 400 stays from set-b, never seen, and let the memory make one pass over a pool of set-a stays first (Table 10). Each doubling of the pool lowers the codelength on the unseen stays by about 0.045 bits, without saturating; PPM falls at the same rate, and the margin stays flat near 0.80. Past one pass, both absolute codelengths worsen while the margin widens (epochs 1, 2, 4 over 800 stays: 3.320, 3.345, 3.431 against PPM 4.125, 4.199, 4.295). The absolute column, not the margin, is the one to read. Transfer here is what counting would also give.

pool (stays)	0	200	400	800	1600	3200
ours	3.716	3.433	3.368	3.320	3.276	3.233
PPM-C o2	4.415	4.235	4.182	4.125	4.074	4.031
margin	0.70	0.80	0.81	0.80	0.80	0.80

Table 10: Bits per event on 400 unseen stays after one pass over a pool.

7.3 A prediction at one position is not a relational question

In-hospital death is fixed at one position, the end of the stay, and it is a function of an order-free summary of the whole stay. Under a five-fold cross-validated linear probe (Table 11), adding bigrams to the count summary lowers it, from 0.803 to 0.770: order carries nothing here beyond counts. The memory’s state pooled over the stay comes within 0.03 of the count summary, and its state at the final tick is 0.71: what the present state holds is recency-weighted, and the question wants the whole stay.

The same question was then asked in the architecture’s own shape: the outcome is an event the world speaks at the end of the stay; the first half of the stays are learned from and the second half read; during a stay the question is put at a quarter, a half and three quarters of the way through (Table 12). The wiped arm is the same memory with the stay erased before each question, and sits at chance: what the memory answers comes from the stay. The online counter is the comparator on the model’s terms — one pass, each stay predicted before it is learned from — and the batch counter differs from it by at most 0.01, so the gap is not an artefact of the online regime.

The end-of-stay figure matches the final-tick probe (0.709 against 0.708): the readout takes out of the present state about what a linear probe does, and what the present state lacks is the

representation	AUROC
count and mean-gap summary of the whole stay (<i>control</i>)	0.803
memory state, max-pooled over the stay	0.777
count summary plus hashed bigrams	0.770
memory state, mean-pooled	0.754
hashed bigrams alone	0.750
memory state at the final tick	0.708
SAPS-I through the same probe	0.635
timing alone, best single feature	0.616

Table 11: Five-fold cross-validated linear probe, the same probe and regularisation sweep for every row.

asked at	memory	stay wiped	counter, online	counter, batch	answer changes
25%	0.564	0.496	0.741	0.751	0.61
50%	0.643	0.496	0.765	0.764	0.49
75%	0.675	0.496	0.775	0.781	0.41
end	0.709	0.496	0.794	0.796	0.28

Table 12: AUROC on the second half of the stays. The last column is the fraction of questions whose answer changed during the silence after the ask.

whole stay. The charge on the spoken outcome, 0.607 bits, is above the base-rate entropy of 0.590, and answers still change during the silence: the step of the mechanism chases a label this noisy, as Table 8 predicts. We did not adopt a smaller step for this source, for the reason given in §6.3.

What the source can and cannot say is now plain. It shows a memory that accumulates over a million events where a bounded context cannot, and that answers a question about a stay from the stay. It cannot show organisation beyond counting, because its one outcome is a count, and it cannot show response, because nothing the model says changes what the patient does next.

8 Explorations on top of the mechanism

Each component in this section was built on the mechanism of §3, measured, and left in the implementation as a switch that is off. Each is reported with what it maintained, what it cost, and why it is not part of the mechanism. The measurements are the ones already introduced: the facts of Table 2, the synthetic closeout of §6 (two seeds), and the clinical codelength of §7. None of these components was rejected for raising the charge alone, and none would have been adopted for lowering it alone.

8.1 The facts each learner maintains

Table 13 orders the learners by inclusion. Every exploration above the rule is *equal* to the mechanism in this order except the episode trace, which is strictly above it by F11. Below the rule, the order among the learners that fail something is partial: 8 incomparable pairs, width 4, and order dimension between 2 and 4, so no single number represents it faithfully [Dushnik and Miller, 1941]. Two entries show the limits of the list itself. Whether a fact belongs on it is a design decision the order does not make: the episode trace is above the mechanism only because F11 is on the list. And the order can name a missing fact only if that fact is on the list: the explorations it calls equal differ elsewhere, below.

learner	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
the mechanism	✓	✓	✓	✓	✓	✓	✓	✓ .18	✓	✓ 4.87	.12
read-back check on	✓	✓	✓	✓	✓	✓	✓	✓ .17	✓	✓ 4.87	.12
echo gate on	✓	✓	✓	✓	✓	✓	✓	✓ .24	✓ .001	✓ 4.84	.09
situated memory on	✓	✓	✓	✓	✓	✓	✓	✓ .19	✓	✓ 4.87	.15
episode trace on	✓	✓	✓	✓	✓	✓	✓	✓ .18	✓	✓ 4.87	✓ .75
graph kept, anchor 0	✓	✓	✓	✓	✓	✓	✓	✓ .17	✓	✓ 5.10	.10
anchor 0	✓	✓	✓	✓	✓	✓	✓	✓ .24	✓	✓ 4.64	.11
no self block	✓	✓	✓	✓	✓	✓	✓	.79	✓	✓	.07
sixteen negatives	✓	✓	14.5	✓	✓	✓	✓	2.00	✓	✓	.12
no codebook term	✓	✓	✓	✓	✓	✓	✓	✓	✓	6.00	.12
step 0.5	✓	6.73	✓	✓	✓	✓	✓	✓ .07	✓	✓	.36
rows off (<i>control</i>)	4.91	✓	✓	✓	.50	.61	.61	7.11	✓	✓	.00

Table 13: A tick means the fact is maintained; a number without a tick is the measured value that failed its threshold. Selected passing values are given where they differ between learners. Below the rule, learners that each fail a fact the mechanism maintains.

8.2 The routed operator graph

An earlier design passed the state through a learned transform chosen by content from a small-world graph. With the graph kept, every fact is maintained as without it, and the walk alone answers a relation slightly worse (5.10 against 4.64 bits, the same configuration otherwise). On the synthetic stream every reading falls (Latin $0.486 \rightarrow 0.455$, walk support $0.379 \rightarrow 0.327$, answer bits $3.98 \rightarrow 4.14$). On the clinical stream the charge rises from 3.319 to 3.366 bits per event, and the margin on new contexts falls from 0.26 to 0.15. We removed it because it maintains no fact the mechanism lacks and costs on every other reading, not because of the charge. A stream that reported which graph node the state last passed through belonged to it; with the graph removed it reported a constant, and when repaired to report the last bank written it raised the charge (3.343) and changed nothing on the synthetic stream. It is off.

8.3 Anchoring the state to the background

The mechanism pulls p toward the background bands at each tick (anchor 0.35). Without the pull every fact is maintained; the walk alone answers a relation better (4.64 against 4.87 bits) and a new pair slightly worse (0.24 against 0.18), and the synthetic readings are within seed variation of the mechanism’s. The facts do not decide this pair, and we keep the anchor for the reason it was introduced: a state with nothing tying it to elapsed time drifts wherever its own walk takes it.

8.4 A read-back check on the emission

Before an answer is said, the check unbinds it back through the key it was retrieved under and refuses it if it does not return. It is equal to the mechanism in the order of facts; on the synthetic stream it lowers Latin from 0.486 to 0.455 and raises answer bits from 3.98 to 4.08. An earlier measurement of it on the clinical probe was its main support, and that measurement was on a question §7 shows to be an order-free count. It is off.

8.5 A gate on silent hops that return to what was just heard

A silent hop can collapse onto the token just heard, which lets go of what preceded it. The gate refuses such hops. It was built on an accuracy reading of F9 that proper scores reversed (§5): the silence improves an ungrounded answer rather than worsening it. With the gate the

improvement is lost (+0.001 against -0.058 , still inside the threshold), and composition of a new pair worsens (0.24 against 0.18). When the gate was first built, an explanation of its gain was tested — that the gate preserves order information through long silences — and was not supported: on a history question with silences of eleven ticks it moved the answer by about one standard error. A change we cannot explain is not adopted for its charge. It is off.

8.6 What was said, bound into what followed

The mechanism binds what it said into the key of what the world did next (Eq. 8). On every source in this paper this is neutral: the closeout without it is within 0.005 of the mechanism on every reading, the facts are identical, and on the clinical stream 131358 of 1.86M triples carry a said token, at no measurable cost. It is in the mechanism because it is the only path by which a consequence of the model’s own response can be learned, and no source here has consequences (§10). Its first version wrote the said token unmarked into the world’s key space, where it collided with the key of a silent walk and broke a finished walk’s hold on its answer (Table 15); the said token now carries a self marker and an identity space of its own.

8.7 An exact trace per context

A dialogue source (Appendix A) showed that the mechanism has nowhere to read “what this context bound to X ” from: the superposed store holds what X is usually bound to. An exact trace, keyed by the pair and overwritten on write, answers the question: F11 rises from 0.12 to 0.75, and nothing else changes. It is off because it is a transcript. The design’s commitment about context (§2) is that the present is a context, not a record, and an exact overwrite-on-write store is the record that commitment refuses. The fact it maintains is real; the question it raises is whether that fact belongs to memory of this kind or to a separate faculty.

8.8 Situated memory

The question the episode trace raised has a form that does not need a transcript: bind each triple to the situation it happened in, and read with the situation reinstated, as the temporal context model does [Howard and Kahana, 2002]. The situation is the first level of the world’s own background cascade, unitarised so that binding by it is invertible; the situated triples go into a superposition of their own, beside the main store.

None of the real sources has situations that recur, so the source for it is built to have them: four or eight backgrounds, eight frequent pairs whose successor depends on the situation, and 32 rare pairs, visited in random order with gaps between visits. Recall is scored as the cosine of the unbound value with the right successor, situated against plain (Table 14).

recall of the situation’s successor	4 situations		8 situations	
	seed 7	seed 8	seed 7	seed 8
later in a visit, situated	0.804	0.745	0.517	0.499
later in a visit, plain	0.263	0.252	0.120	0.126
first event of a visit	0.365	0.328	0.213	0.167
after a gap of 0–1 visits	0.870	0.738	0.602	0.448
after a gap of 10 or more	0.781	0.684	0.489	0.490
rare pair seen once	0.204	0.220	0.097	0.155
rare pair seen twice	0.267	0.295	0.208	0.245
rare pair seen 3–5 times	0.448	0.468	0.426	0.510
rare pair seen 6 or more	0.735	0.798	0.816	0.789

Table 14: Cosine of the recalled value with the correct successor, $d = 512$.

Three things hold on both seeds. Situated recall is three to four times plain recall on the frequent pairs, whose successor depends on the situation. It is robust to the gap: after ten or more intervening visits it is close to what it is after none. And a situation that is reassigned — the successors redrawn every eight visits — is relearned: recall of the new successor passes the old one on the first visit after the change (0.42 against 0.23 and 0.41 against 0.18) and on seed 7 the charge on it falls from 1.58 to 0.41 bits.

What does not hold is one-shot recall. A rare pair seen once in a situation is recalled at 0.10–0.22, barely above plain recall (0.16), and reaches the frequent pairs’ level only after six exposures. This is the fan effect [Anderson, 1974]: every other situation’s writes of the same pair load the same bank, and a single exposure is one trace among all of them. The remedy biology uses is pattern separation, sending similar inputs to distinct codes [Yassa and Stark, 2011]. We tried its hard form, selecting a sub-bank by sign bits of the situation. It made recall worse (later in a visit 0.80 $\rightarrow$ 0.70; after a long gap 0.78 $\rightarrow$ 0.53), because the situations of this source are not stable enough to be separated: two visits to one situation are only 0.56 alike, against 0.06–0.09 for two different situations, so the bits disagree between visits to the same place.

On the static sources the situated store is neutral in the order of facts and on the synthetic stream, and it costs 0.034 bits per event on the clinical stream (3.353 against 3.319), where there are no situations to recur. The charge moving the wrong way is not why it is off — it is off because no source here has what it needs. An environment with persistent surroundings is the test it waits for (§10).

8.9 Other switches

Cleanup restricted to tokens actually heard, and the codebook term of the write restricted to them, were hypotheses for the difference in Table 5; both leave the synthetic readings where they were, and are off. Step rules that shrink the step as a row settles are reported in §6.6; neither is adopted.

9 Method: failures of one shape

Most of what went wrong in this project went wrong in the same way: a mechanism measured inert, or a number read as a result, and the cause lay between the mechanism and the reading.

The discipline this forces, stated because we expect it to generalise to any system of this shape:

1. Any change on the path of a load-bearing mechanism gets that mechanism’s off-control *in the same sitting*. If the full arm and the off arm are close, stop.
2. A terminal metric cannot debug a multi-step chain. Four-second unit tests found errors that ten-minute end-to-end runs could not.
3. Bit-identical arms are more alarming than bad numbers.
4. Any hand-set constant must be back-solved from its own units.
5. Suspect the data before the mechanism, and the instrument before either.
6. Before reading a number as the quantity wanted, ask what else it could be. Four readings in this project were something else: a trend on a subset that was the learning phase, a class that was a phase, a descent in progress read as a floor, and accuracy on answers with no information in them.
7. A synthetic source built with symmetries — uniform marginals, no shared structure — hides every defect those symmetries make invisible. Each real source found at least one.

symptom	cause	what found it
self-stream ablation reads zero	feature blocks wider than rows; background bands silently truncated	per-block alignment assertion
near-miss curve flat	the write path was dead code; reads and writes landed on the same perturbed edge	call-graph search
every accuracy collapses	a decay constant applied <i>per tick</i> , leaving 1.6% of the binding by the time it was read	the mechanism’s own off-control
six arms bit-identical	the retrieval never reached the emission	the bit-identity itself
cleanup succeeds 97.6% and is always wrong	the threshold sat <i>below</i> the codebook noise floor	measuring four numbers directly
the readout charges a uniform code on real data	scores clamped rather than shifted by their maximum; the synthetic source had uniform marginals, so it hid the defect	an i.i.d. stream with a known charge
rare tokens charged near 2^{-50}	the write pushed down only the sixteen highest-ranked tokens	excess split by rarity
the write stream a constant	with the graph removed it still reported the graph’s node	reading what the stream carried
a finished walk leaves its answer	what was said entered the world’s key space	the surplus-silence assertion

Table 15: Each symptom reads as “a mechanism measures inert” or “a number is a result”. Each cause is that the mechanism’s product had no outlet, was erased downstream, or was confused with something else.

Performance belongs in the same list, because where computation is bought with the world’s time it is not merely engineering. The $O(d^2)$ convolution became $O(d \log d)$, which is also what makes the unitary codebook affordable.

10 What static data cannot test, and what would

10.1 Every evaluation here is open-loop

In every experiment of this paper the world’s next event is fixed before the model says anything. Questions put to the model are answered by the world whatever the model answered; the clinical outcome is spoken at the end of a stay whatever the model expected; the synthetic source proceeds on a schedule. Such evaluations test memory — what is stored, what is retrieved, how it degrades — and prediction. They cannot test the response half of the design: what is said, and what the world does because it was said.

This is not a shortcoming of the particular sources. It holds for any recorded stream, because a record is precisely an answer that does not depend on who asks or what was asked before. A learner whose behaviour changes what it subsequently experiences is not described by any record of its inputs, and the order in which it asks shapes what it can learn: along questions that do not commute, what is learned in the limit depends on the strategy of asking. That is the situation a challenge–response learner is built for, and it is the one a replayed file removes. Nor can a replayed file test time as a resource: the model receives exactly the ticks the file specifies however much computation it needed, and is never late.

10.2 What a world must have

The properties below are what the preceding sections needed and could not get. Each is stated as a property of the world, not of a benchmark.

1. *A clock of its own.* The world proceeds while the model is silent; thinking longer costs the chance to act or to notice. Only then is depth bought with silence a trade rather than a free lunch.
2. *Influence without generation.* What the model does changes what it experiences next — where it is, what it can see, what reacts — but the world’s own processes continue regardless. This is what excludes the dark room: a learner cannot make the world quiet by hiding from it.
3. *Laws that recur across situations.* The same regularities hold in many places and objects, so that what is learned here can shorten learning there. Without shared structure, organisation has nothing to organise and transfer is impossible by construction.
4. *Consequences as events.* The world’s reactions are further tokens, charged and written like any other; nothing is a label and nothing is a score.
5. *Partial observability.* Where to look and what to do decide what is learned, so that the model’s own strategy is part of the experiment.
6. *Situations that persist.* An environment with stable surroundings, so that “being here again” is recognisable. The situated store of §8 was limited by exactly this: our synthetic situations were random samples of tokens and two visits to one of them were only 0.56 alike.

A simulated physical environment has all six by default. Its laws are known to the experimenter because the experimenter wrote them, which gives a reference for reading what the learner has organised without handing the learner any answer; it contains no reward unless one is added; and its processes run whether or not anything attends to them. A discrete lattice physics with objects, collisions and gravity keeps the laws while avoiding pixels and continuous control, so that the difficulty lands on memory and response rather than on perception. Richer simulations, and eventually physical robots, are the later rungs of the same ladder.

10.3 Why anything would move: drives and reflexes

An organism is not without reward and punishment; without them it would not move. The design’s commitment is narrower: valence must not reach the write. We read that as three constraints.

1. *Valence is experienced as events.* Pain, satiety, cold and fatigue are tokens from the body — part of the world — charged and written like anything else. Memory comes to hold what tends to precede pain because pain is an event, not because it is bad; the write rule does not know the difference. How deeply a painful event is written is decided, as for any event, by how surprising it was.
2. *Valence acts on expression, not on storage.* Among the responses maturing in parallel, the one to say is chosen with regard to what it is expected to lead to: a response whose walk through memory reaches pain is held back, one that reaches relief is let through. Looking further ahead is walking further along the chain, so how far the learner foresees consequences is bought with silence like any other depth. The choice changes what is done now and never what is stored. To keep it from becoming a value function in disguise, only the first valenced event on a walk is consulted, and nothing is summed.

3. *Drives are several and incommensurable.* Hunger, pain and cold do not share a currency. Keeping them apart keeps valence from becoming the single scalar the design refuses, and makes the trade-offs between them a partial order like the one we measure learners by.

A few fixed reflexes — withdrawing from pain, orienting to a sudden event — give behaviour a seed before memory holds any consequence to recall, and drives that rise on their own clock, like hunger, guarantee that the learner cannot stay still for long. Designing these, and designing the world, are engineering problems; the constraints above are what the mechanism asks of them. The marked keys of Equation (8) are the memory such a learner would learn its own consequences from.

10.4 What would be measured there

Nothing in such a world has a fixed answer, and nothing should be scored by one. The instruments of §5 carry over as facts about behaviour:

- whether the codelength of the world’s reaction to the learner’s own acts falls — a reading of whether it has learned what its acts do;
- whether regularities learned in one region shorten learning in another (acquisition that accelerates, which our sources could not show);
- whether its expectations stabilise, and after how many changes;
- how its own order of looking and acting shapes what it learns;
- whether what it has organised in memory corresponds to the laws the simulator was written with.

10.5 Where it might be useful

We state these as directions the properties point to, not as claims; none has been tested.

- *Long-running embodied agents.* A robot in an unstructured setting meets situations that were not enumerated, receives them once and in order, has no reset and no labels, and has idle time. Continual single-pass writing, a cost per tick that does not grow with history, and computation that deepens in a lull match that shape.
- *Devices with bounded compute.* Sensor nodes and wearables have power budgets and long quiet periods. A learner whose depth is bought with silence spends compute when there is time for it, and a fixed per-tick cost is a fixed energy per event.
- *Personal and assistive systems.* Learning one person’s routines and places without labels, and recognising a situation when it recurs, is the kind of situated, long-range memory §8 began to build and could not finish on a synthetic source.
- *Processes whose timing is information.* In monitoring streams — clinical, industrial, laboratory — when something is measured carries information of its own (§7). Spending the gaps as computation rather than as a symbol to condition on is what the design does with time.
- *Interlocutors in real time.* Conversation is the picture the design started from. A dialogue partner that thinks while the other is speaking and speaks when the lull allows is only testable live, with the other party able to react.

11 Limitations

Open-loop throughout. No result here tests the response half of the design (§10). What is reported is memory and prediction.

Replay cannot test time as a resource. The synthetic silences are authored and the clinical ones are compressed logarithmically and capped. The model is never late; there is no rate. Binding the compute budget to wall-clock time is where depth-from-time becomes a genuine constraint, and it has not been done.

Sources. One synthetic source, built by us, whose regimes share no latent structure, so that acquisition cannot accelerate on it by construction; one real stream, whose one outcome is an order-free summary. Real-data results are single-seed; synthetic results use two seeds.

One learner, one step. The readout’s delta rule is the only learner, and one fixed step cannot serve both deterministic structure (which prefers a large step) and a noisy target (which needs a small one). Separating learning structure from chasing noise needs a quantity other than the error, and we do not have it (§6.6).

Episodic memory is unfinished. The situated store recalls a recurring situation’s own successor across long gaps but not from a single exposure, because every other situation’s writes of the same key load the same bank; hard pattern separation made it worse on our source (§8).

Caution. We do not know what these mechanisms do in a world that answers back, and do not think it can be predicted from here.

12 Related work

Distributed binding. Our store is a superposition read by correlation, in the holographic [Plate, 1995] and tensor-product [Smolensky, 1990] traditions, addressed the way sparse distributed memory addresses [Kanerva, 1988], and surveyed as hyperdimensional computing [Kanerva, 2009, Gayler, 2003, Eliasmith, 2013, Kleyko et al., 2023]. Capacity of such stores is analysed by Plate [1995] and Frady et al. [2018]; our Proposition 2 is the form we needed, coupling capacity to a cleanup codebook of size V . Classical associative memories [Willshaw et al., 1969, Hopfield, 1982, Kohonen, 1972] are the ancestors of the readout.

Learned versus fixed addressing. Differentiable memories [Graves et al., 2014, 2016, Weston et al., 2015] and modern associative layers [Ramsauer et al., 2021] learn where things go. Ours is fixed by construction, following Yang [2026]: an address that no later learning can move is what buys freedom from interference, and the price is retrieval difficulty. Fast-weight schemes [Schmidhuber, 1992, Ba et al., 2016] also write associations at inference time, but inside an autoregressive objective.

Random transforms. The routed operator graph we removed behaved as a random feature expansion [Rahimi and Recht, 2007] feeding a linear readout, which is the regime reservoir computing occupies deliberately [Jaeger, 2001, Maass et al., 2002].

Episodic memory and its biology. The situated store of §8 follows the temporal context model, in which an item is bound to a slowly drifting context and recall reinstates context [Howard and Kahana, 2002], and the complementary-learning-systems account of a fast episodic store beside a slow one [McClelland et al., 1995]. Its failure mode is the fan effect, the slowing and weakening of retrieval as more facts share a cue [Anderson, 1974]; the remedy biology uses is pattern separation in the dentate gyrus, which sends similar inputs to distinct codes [Yassa and Stark, 2011], and our hard, sign-bit version of it did not survive a source whose situations were unstable.

Continual learning. The setting and its interference are classical [McCloskey and Cohen, 1989, French, 1999]; responses include regularisation [Kirkpatrick et al., 2017], replay [Lopez-Paz and Ranzato, 2017], and architectural allocation, surveyed by Parisi et al. [2019]. Our position is the allocation one, taken to its committed form, with no replay and no task boundary.

Compression and prediction. The ledger is prequential [Dawid, 1984, Rissanen, 1978] in the description-length reading of learning [Blier and Ollivier, 2018, Grünwald, 2007], and the identification of compression with understanding is argued at length by Solomonoff [1964] and Hutter [2005]. What we do not take is the objective: there is no reward here [Sutton and Barto, 2018, Silver et al., 2021], and codelength is read as an instrument.

Embodiment and the dark room. That behaviour influences experience without generating it, and that this is what defeats the dark-room objection, is argued in the free-energy literature and its critics [Friston, 2010, Friston et al., 2012, Clark, 2013]; that intelligence should be judged in a world rather than on a corpus is the embodied tradition’s position [Brooks, 1991, LeCun, 2022].

Autoregressive systems can do these tasks. A long-context model can solve retrieval of this shape by induction [Olsson et al., 2022], and state-space models [Gu and Dao, 2023] carry a running state rather than a window. Neither can spend a lull: computation per token is fixed, and that is the axis on which this work differs.

13 Conclusion

We built a challenge–response memory in which binding and unbinding are the two halves of one tick, selected by whether the world spoke, in which several partial responses mature in parallel while the world is quiet, and in which what the model says is heard back and bound to what followed. In isolation the mechanism does what the design says: one link per quiet tick, a halt at the end of a chain, surplus silence harmless. We derived the operating region of its store, showed that binding inverts exactly only for unitary codebooks, and showed that locality-sensitive addressing makes a near miss undetectable by construction.

On static sources the mechanism maintains every fact on our list but one, and that one belongs to a transcript the design refuses. On a synthetic stream its load-bearing parts carry what they should; on a real stream it accumulates over a million events where a bounded context cannot, and loses to a counter only where the delta rule has not yet arrived. What was built on top of it and left out is reported with the reason, and in no case was the reason the charge. Several parts of the design read as neutral or negative on these sources — hearing itself, binding what it said, a deep background, a situated store — and each is the part a static source cannot exercise, because on a static source nothing the model says changes anything.

That is the claim we make and the one we cannot yet test. The design is for a learner in a world that has a clock and that answers back. Where it should be judged is in such a world,

simulated or physical, and we expect that judgement to be substantially harder than anything reported here.

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A Screening sources

Each source was screened with memoryless judges before a model was run on it (§5). None of these numbers involves the model.

Concept-drift streams. Electricity is sampled every thirty minutes without exception (coefficient of variation of the gaps 0.000), and predicting the previous label scores 0.853 against 0.752 for naive Bayes. The stream has no inter-arrival structure for silence to spend, and a trivial judge dominates: it cannot show anything about a memory.

Intensive-care streams. PhysioNet/CinC Challenge 2012 [Silva et al., 2012], set-a. Using only *when* measurements were taken and never their values, in-hospital death is predicted at AUROC 0.616 (the number of measurements in the last six hours), against 0.637 for SAPS-I and 0.626 for SOFA, which are computed from the values: the observation process carries information of its own. The outcome itself is an order-free summary (§7), so the source suits next-event structure and does not suit questions about order.

A process log. BPI Challenge 2012 [van Dongen, 2012], 13087 loan-application cases. Structural questions about a case so far — has an offer been sent, is the latest offer event a cancellation — are answered at 0.95 or more by the last two events or by counts. Questions about elapsed time leave room (over seven days since the offer was sent: counts with elapsed time 0.897), but they reduce to keeping a clock. Unsuitable.

Task-oriented dialogue. MultiWOZ 2.2 [Budzianowski et al., 2018, Zang et al., 2020], 10437 dialogues and about 6×10^5 questions “what does the user currently want for slot *s*”, asked after every user turn. The rule “the last value the user gave for *s*” answers 0.908; restricted to the last three turn pairs, 0.677; the most frequently mentioned value, 0.579; the last value either speaker gave, 0.582. The question passes the screen — it needs long-range retrieval by domain and slot,

keeping who spoke — and is still unsuitable for this design: about nine questions in ten are literal, a dictionary keyed by speaker, domain and slot answers those at 0.99, and no memory can do better than match it there. What remains, mostly an offer accepted with “yes”, is too small and too conventional to test organisation. It is the source that showed the mechanism had nowhere to read “what this context bound to X ” from, which led to the explorations of §8.

B Reproducibility

The implementation is 20,250 lines of Rust with no dependencies, deterministic under a counter-based generator with fixed-order float reductions. It carries 36 structural assertions and 6 source self-tests that run on every build; two identical runs agree bit for bit. Constraints A1–A5 and each mechanism claim of §4 are asserted by a named test. Every result in this paper was produced by one build of the mechanism as described, commit [7927848](#); the commands, seeds and raw outputs are listed with the code.